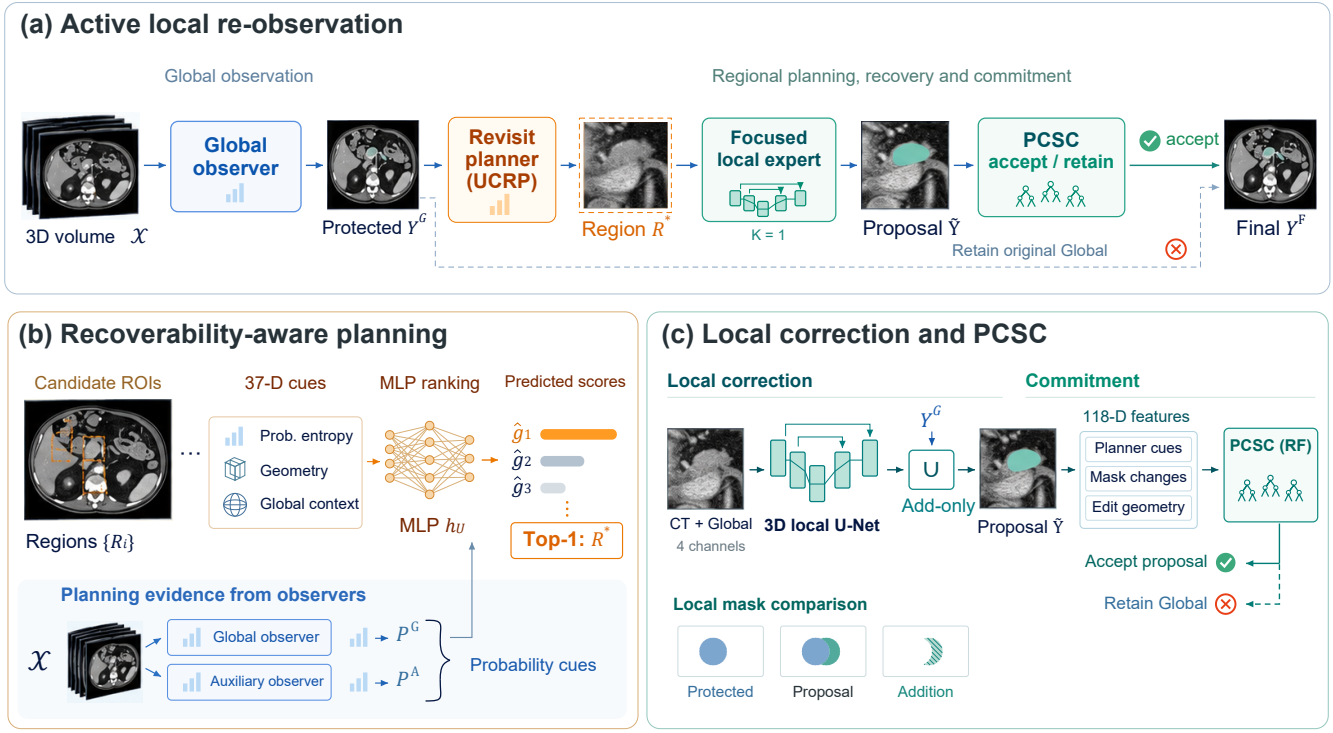

(a) Active local re-observation
Global observation
Regional planning, recovery and commitment
3D volume $\mathcal{X}$
Global observer
Protected Y^G
Revisit planner (UCRP)
Region R^*
Focused local expert
K = 1
Proposal $\tilde{Y}$
PCSC accept / retain
accept
Final Y^F
Retain original Global
(b) Recoverability-aware planning
Candidate ROIs
37-D cues
MLP ranking
Predicted scores
Regions $\{R_i\}$
Prob. entropy
Geometry
Global context
MLP h_U
$\hat{g}_1$
$\hat{g}_2$
$\hat{g}_3$
Top-1: R^*
Planning evidence from observers
$\mathcal{X}$
Global observer
Auxiliary observer
P^G
P^A
Probability cues
(c) Local correction and PCSC
Local correction
Commitment
CT + Global
4 channels
3D local U-Net
Y^G
U
Add-only
Proposal $\tilde{Y}$
118-D features
Planner cues
Mask changes
Edit geometry
PCSC (RF)
Accept proposal
Retain Global
Local mask comparison
Protected
Proposal
Addition